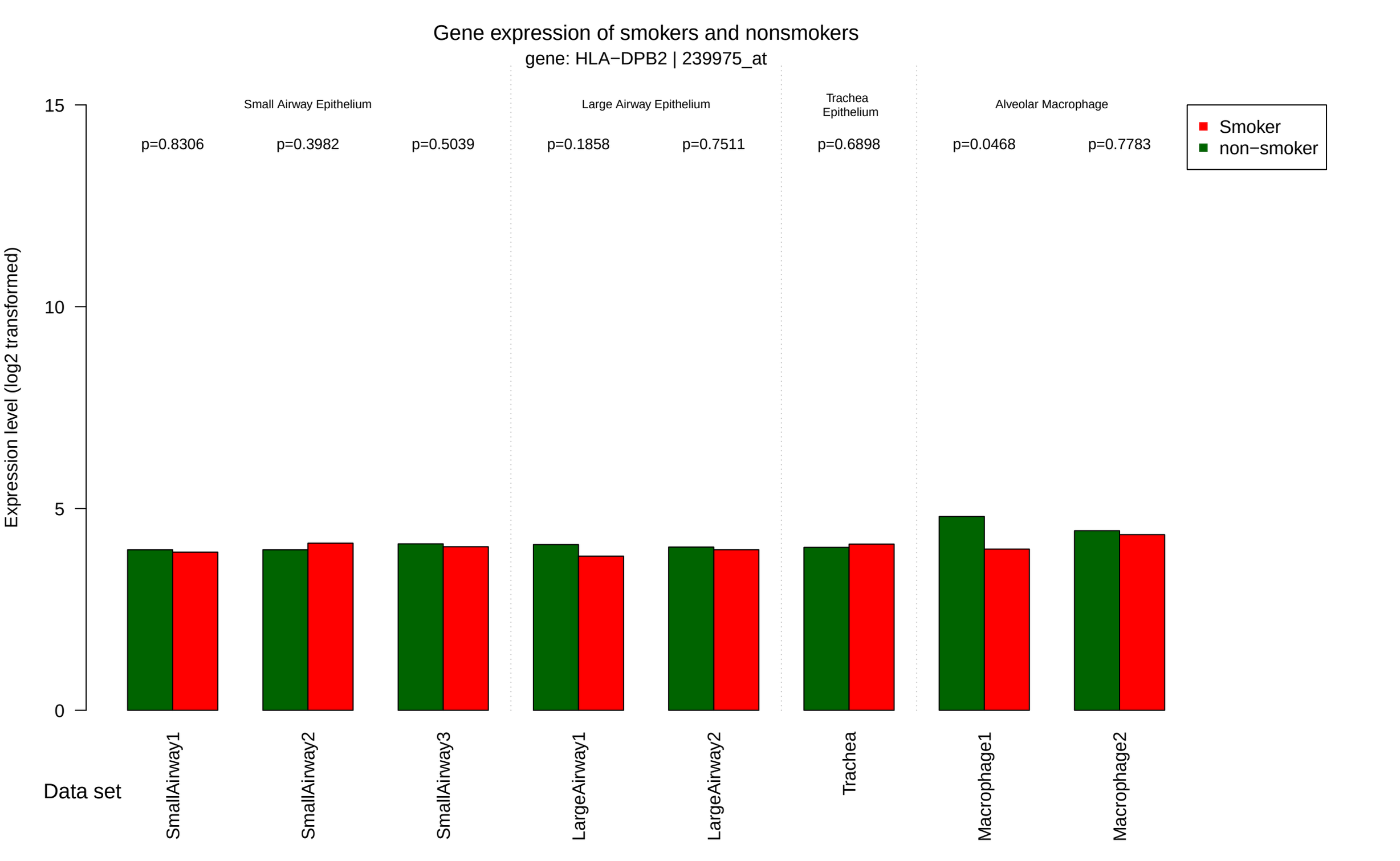
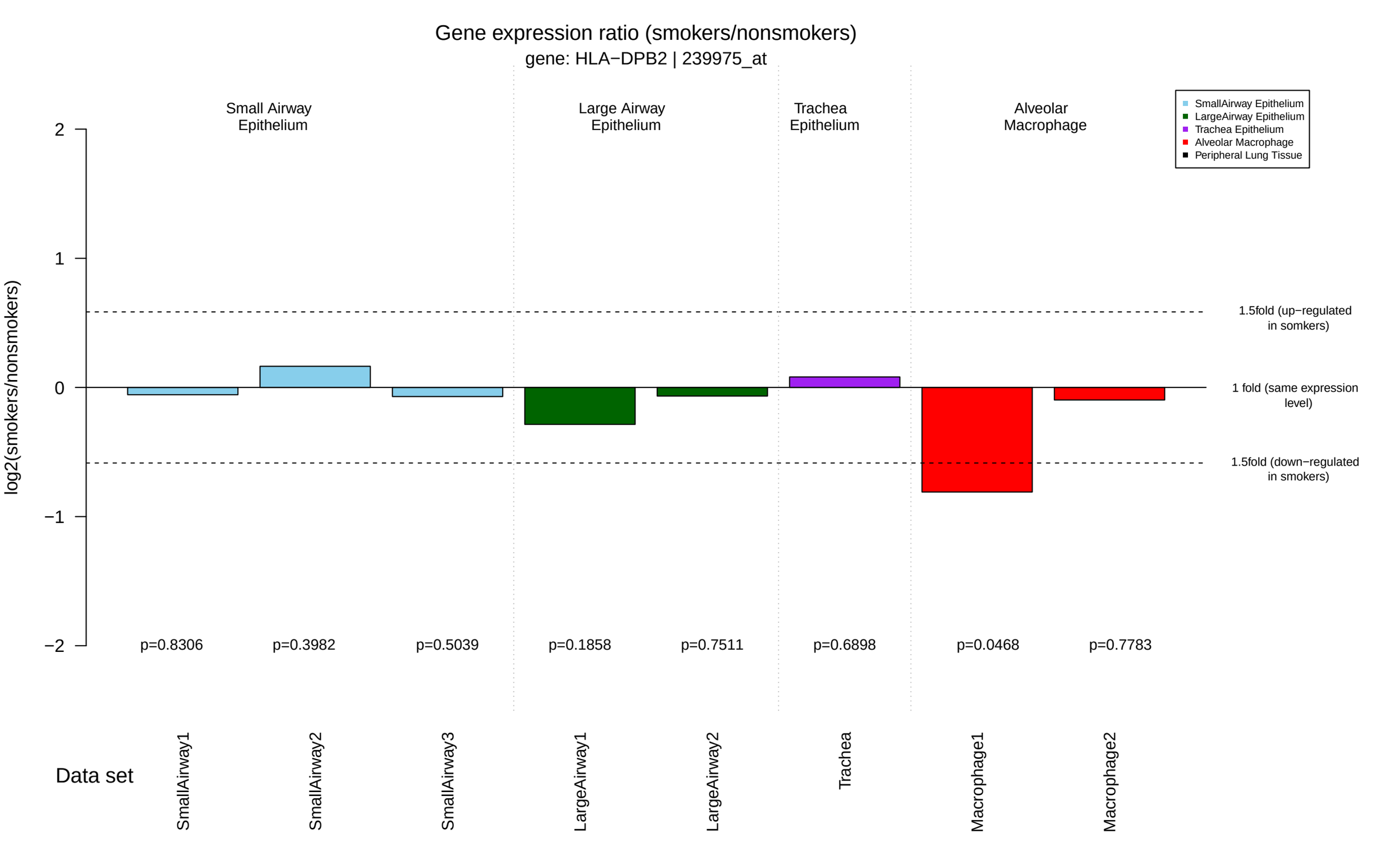
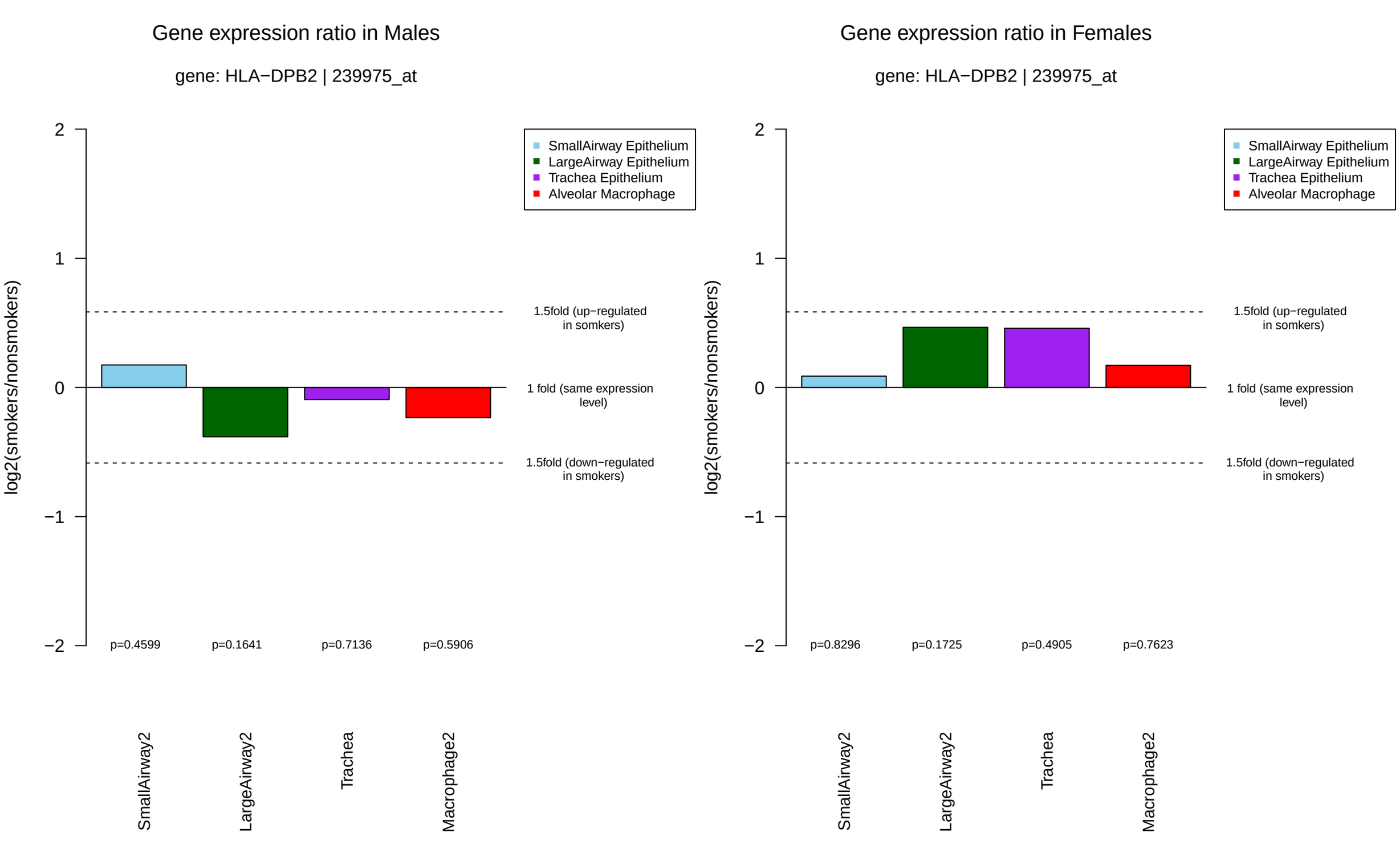
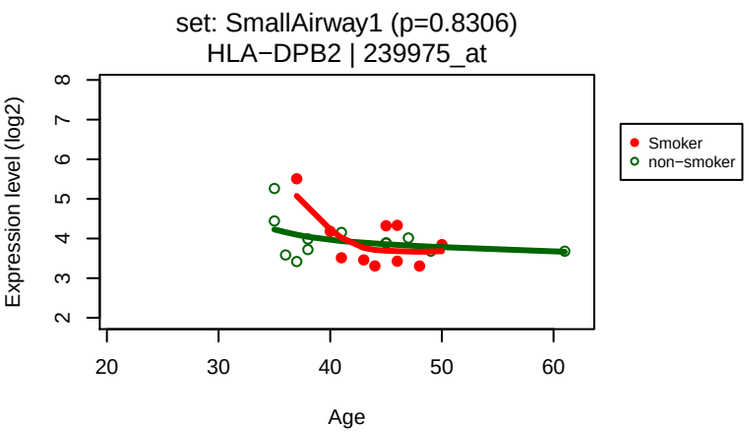
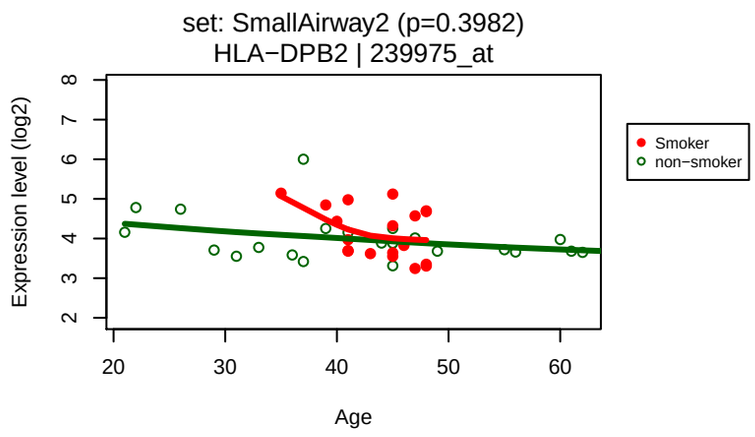
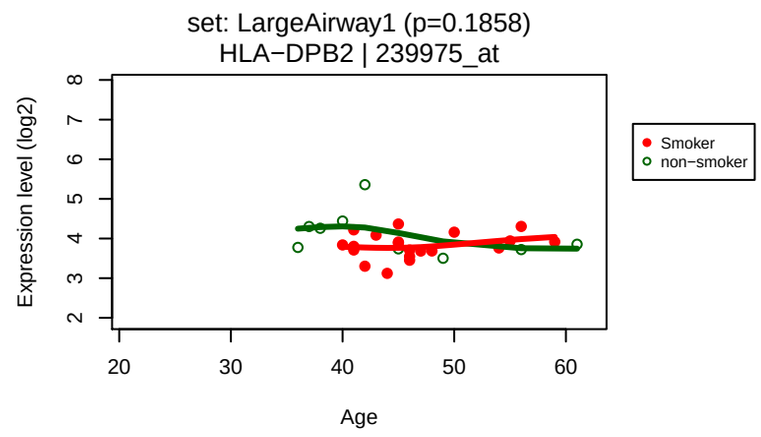

Gene expression of smokers and nonsmokers
gene: HLA−DPB2 | 239975_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8306
p=0.3982
p=0.5039
p=0.1858
p=0.7511
p=0.6898
p=0.0468
p=0.7783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

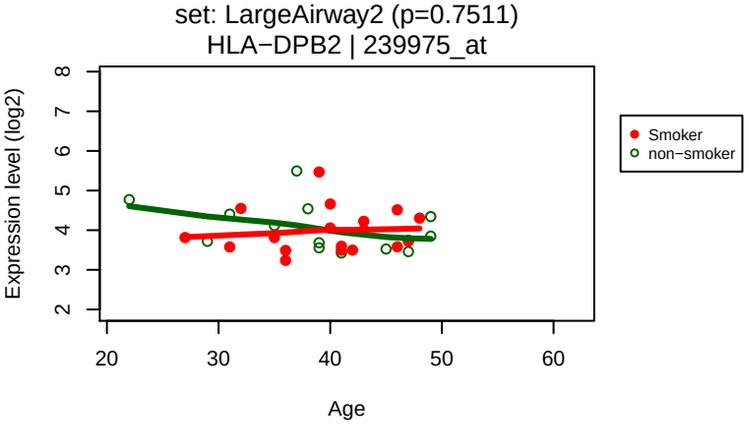

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DPB2 | 239975_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8306
p=0.3982
p=0.5039
p=0.1858
p=0.7511
p=0.6898
p=0.0468
p=0.7783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

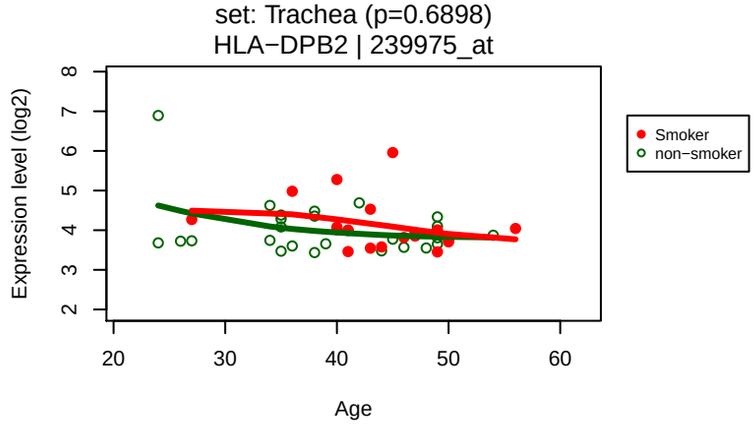

Gene expression ratio in Males
gene: HLA−DPB2 | 239975_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4599
p=0.1641
p=0.7136
p=0.5906
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DPB2 | 239975_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8296
p=0.1725
p=0.4905
p=0.7623
SmallAirway2
LargeAirway2
Trachea
Macrophage2

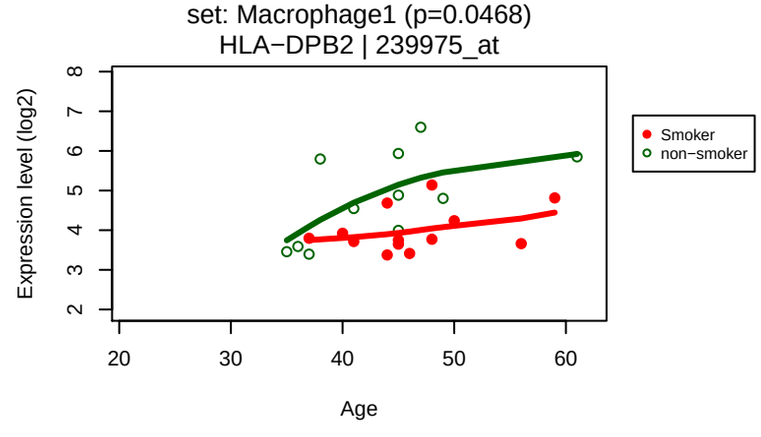

set: SmallAirway1 (p=0.8306)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker

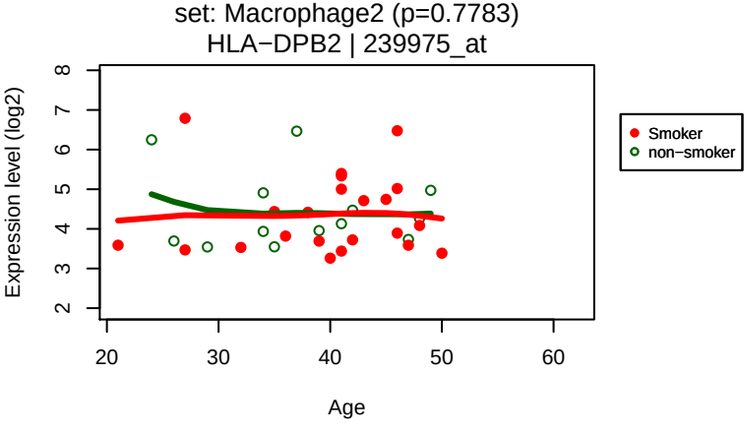

set: SmallAirway2 (p=0.3982)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker

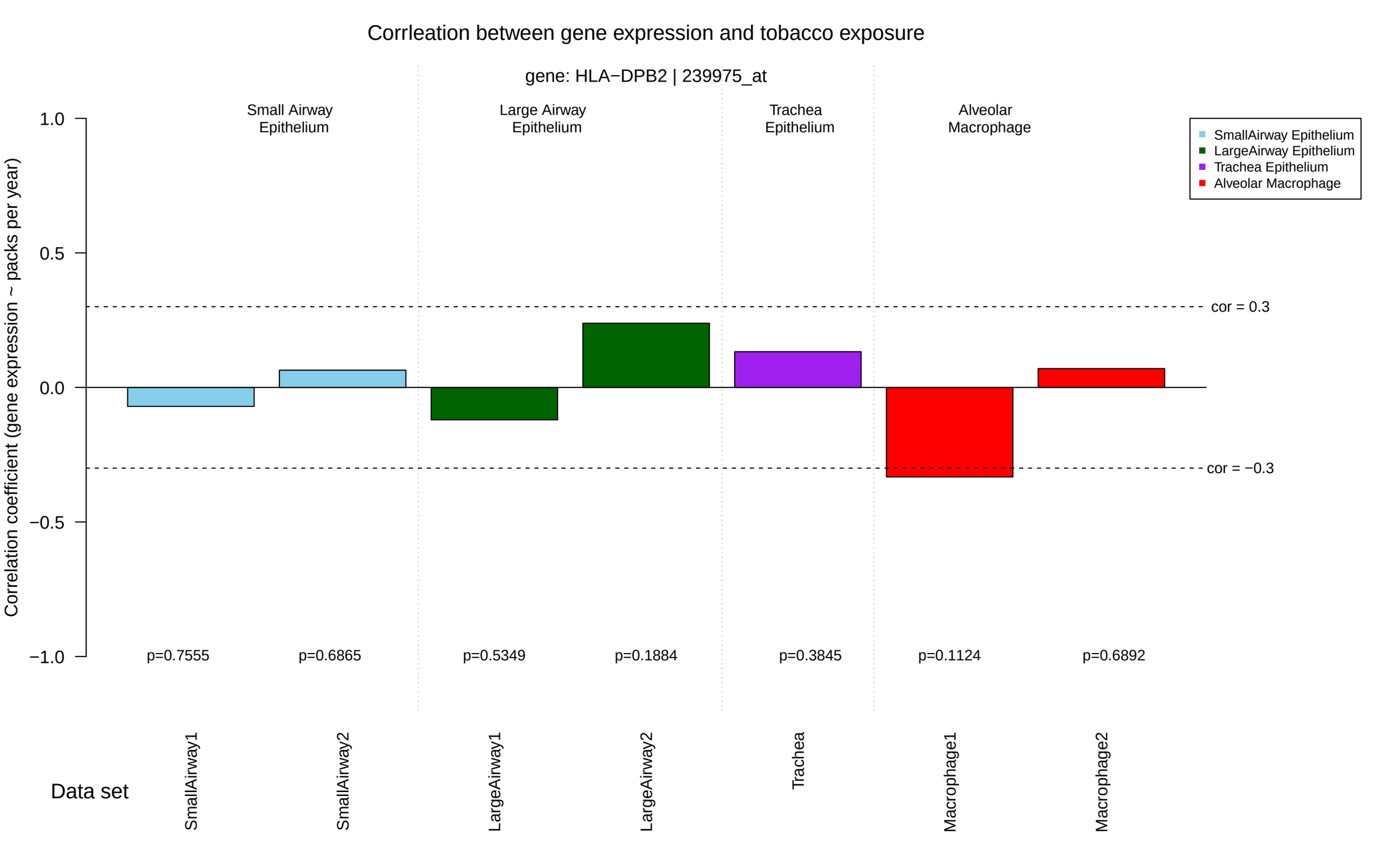

set: LargeAirway1 (p=0.1858)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7511)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6898)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0468)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7783)
HLA−DPB2 | 239975_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DPB2 | 239975_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7555
p=0.6865
p=0.5349
p=0.1884
p=0.3845
p=0.1124
p=0.6892
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2